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Superconductivity of Tantalum-Rhenium Alloys

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The new measurement of the specific heat of bcc 5d transition metal alloys of Ta-Re series containing up to 40 atomic % of Re has been made and their superconducting behaviours of the alloys including up to 20 atomic % of Re have been studied. By making use of McMillan's theoretical treatment, the electron-phonon coupling constant and bare electronic density of states were deduced. The electron-phonon coupling constant, as a function of the number of electrons per atom, behaves in the same way as the density of states. This result suggests that the coupling constant is proportional to the bare density of states. The present results of the density of states were compared with the theoretical ones which were deduced by Mattheiss for Ta and W.

§1. Introduction

It is known that the electronic density of states and the superconducting properties of transition metal alloys, for example, the superconducting transition temperature, vary remarkably with the number of valence electrons. It is interesting to clarify the correlation between the occurrence of superconductivity and the electronic band structure of bcc 5d transition metal alloys. In this paper we wish to present the new measurement of the specific heat and superconducting properties of bcc 5d transition metal alloys Ta-Re series. The experimental results show that the superconducting transition temperatures and electronic specific heat coefficients decrease as electrons-to-atom ratio increases from the valence of 5. As the several data of bcc 5d series Hf-Ta, Ta-W, and W-Re between the valence of 4.7 and 6.25 have been already reported by Bucher *et al.*,[1] the present results on Ta-Re are compared with them. Using McMillan's theory,[2] the electron-phonon coupling constants and the band structure density of states are derived. The band structure density of states of Ta-Re alloys are compared with the theoretical density of states calculated by Mattheiss for Ta and W.[3]

§2. Experimental Procedure

Ta-Re alloy has a wide solid solution range of bcc phase from pure Ta side to 46 atomic % Re.[4] Specific heats for ten specimens were measured and their results are summarized in Table I. Powders of 99.9% pure Ta and Re were mixed in an appropriate concentration and pressed into tablets. These tablets were melted into a button form with an electron beam furnace under the vacua of 10^{-6} mmHg turning upside down several times to ensure spatial homogeneity. The measurement was done making use of those buttons. The melting temperatures for both Ta and Re are similar and their vapor pressures near the melting points are almost equal and moreover the specimens were melted in the form of tablets pressed from well-mixed powder. The deviation of concentration from nominal fraction upon melting was negligible. The nominal concentration was used in the analysis of the experimental results.

The measurement of specific heat was made by the standard adiabatic heat pulse technique. In order to obtain fairly low temperatures, a He^3 calorimeter was used, which is shown in Fig. 1. A He^3 bath contains 4 cubic centimeter of liquid. One of two tubes for He^3 bath was used as the He^3 pumping line and the other was connected to a manometer, to measure the vapor pressure of the He^3 liquid. The sample, addenda with a heater, and thermometer were suspended with nylon threads. The sample was sticked to the addenda with GE 7031 varnish and cooled by means of a mechanical heat switch which was operated externally through a stainless steel wire and bellows. For alloy samples having the transition temperature below 4.2°K, an Allen Bradley 10 ohm carbon resistor was used as a thermometer, and for the Ta sample, a Honeywell calibrated germanium resistor was adopted. Using a Nb-Zr superconducting wire of 0.1 mm diameter and 100 cm long, the thermometer and heater were connected to the outside equipments through a hermetic seal on the top of the can. Although the temperature of the He^3 bath decreased to 0.4°K, the measurement of the specific

Table I. Parameters of investigated specimens. n: number of electrons per atom, γ: electronic specific heat coefficient, Θ: Debye temperature, T_c: superconducting transition temperature, $H_c(0)$: critical field at 0°K, $(\mathrm{d}H_c/\mathrm{d}T)_{T=T_c}$: slope of the critical field at $T=T_c$, $\Delta C/\gamma T_c$: measured reduced jump in zero field specific heat, *: reduced jump calculated with the relationship $\Delta C/\gamma T_c = (\mathrm{d}H_c/\mathrm{d}T)^2/4\pi\gamma$, $2\varepsilon_0/kT_c$: reduced superconducting energy gap deduced by entropy consideration, **: reduced energy gap obtained by the Toxen relationship $2\varepsilon_0/kT_c = T_c/H_c(0)\cdot(\mathrm{d}H_c/\mathrm{d}T)_{T=T_c}$, a and b: reduced superconducting state electronic specific heat $C_{es}/\gamma T_c$ is expressed as $C_{es}/\gamma T_c = a \exp(-bT_c/T)$, λ: electron phonon coupling constant, and $N_0(0)$: bare density of states at the Fermi energy. Parentheses mean interpolated values.

Sample	n Electron/atom	γ mJ/mol·°K²	Θ °K	T_c °K	$H_c(0)$ gauss	$\frac{\mathrm{d}H_c}{\mathrm{d}T}$ $T=T_c$ gauss/°K	$\frac{\Delta C}{\gamma T_c}$	* $\frac{\Delta C}{\gamma T_c}$	$\frac{2\varepsilon_0}{kT_c}$	** $\frac{2\varepsilon_0}{kT_c}$	a	b	λ	$N_0(0)$ states/eV·atom
Ta	5	6.15	250	4.463	831	334	1.60	1.58	3.59	3.59	8.06	1.47	0.62	0.80
Ta 0.975 Re 0.025	5.05	5.70	261	3.458	613	311	1.46	1.53	3.54	3.51	8.60	1.47	0.56	0.77
Ta 0.95 Re 0.05	5.1	5.12	277	2.77	460	296	1.48	1.47	3.48	3.56	8.85	1.46	0.52	0.72
Ta 0.925 Re 0.075	5.15	5.05	285	2.08	342	289	1.40	1.41	3.47	3.51	9.47	1.45	0.48	0.72
Ta 0.9 Re 0.1	5.2	4.10	296	1.49	232	261	1.40	1.41	3.40	3.35			0.45	0.60
Ta 0.85 Re 0.15	5.3	3.75	307	0.75									0.39	0.57
Ta 0.8 Re 0.2	5.4	3.00	317	0.21									0.34	0.48
Ta 0.75 Re 0.25	5.5	2.42	330	<0.06									(0.29)	(0.40)
Ta 0.7 Re 0.3	5.6	1.90	345	<0.06									(0.23)	(0.32)
Ta 0.6 Re 0.4	5.8	1.00	361										(0.13)	(0.19)

heat was only made above 0.5°K at best, due to the temperature rise on releasing the mechanical heat switch. After every run of the specific heat measurement, the thermometer was calibrated against the vapor pressures of liquid He^4 1958 scale[5] and liquid He^3 1962 scale.[6] Below 2°K, He^3 exchange gas was supplied to the specimen vacuum chamber, and the vapor pressure of the liquid He^3 bath was measured with oil and mercury manometers. Above 2°K, He^3 gas in the chamber was pumped out sufficiently, and the vapor pressure of liquid He^4, which was instead supplied to the chamber, was read with a mercury manometer. Two series of calibrations using He^3 and He^4 vapor pressures were connected smoothly. The temperature, T, of the carbon resistor or the germanium resistor was expressed as a function of the thermometer resistance R with an accuracy of ± 0.03°K as follows,

$$\frac{1}{T} = \sum_{n=1}^{7} A_n (\ln R)^{n-4}\ , \qquad (1)$$

where A_n is a constant.* The heat capacity of the addenda alone was measured in a separate run. To obtain the heat capacity of the net specimen alone, it was necessary to subtract the heat capacity of the addenda from the total heat capacity of the specimen including the addenda. The measurement of the specific heat of 99.999% pure copper using this apparatus yielded electronic specific heat coefficient $\gamma = 0.699 \pm 0.003\,mJ/$°K²·mol and Debye temperature $\Theta = 342 \pm 1$°K. The experimental results of the molar heat capacity divided by temperature versus square of temperature for copper are shown in Fig. 2. These results were consistent with the data of many authors, for example, Shen's results $\gamma = 0.696\,mJ/$°K²·mol and $\Theta = 343$°K.[7]

For alloy specimens contained more than 10 atomic % Re, the superconducting transition tem-

* The calculation of eq. (1) and the specific heat were made using the HITAC 5020 computer, The Computer Center, The University of Tokyo.

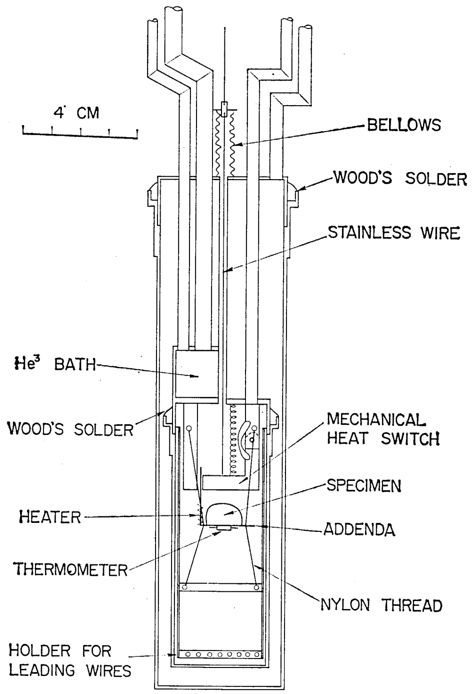

Fig. 1. He^3 calorimeter. Leading wires and hermetic seals are not shown.

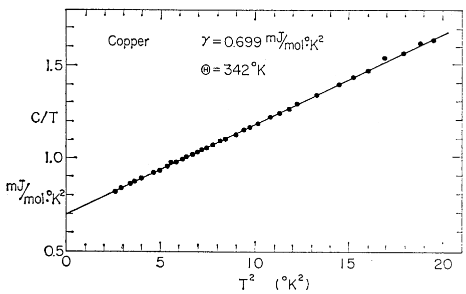

Fig. 2. Molar heat capacity divided by temperature versus square of temperature for copper.

perature could not be determined from the data of the heat capacity alone, because of the fairly low transition temperature. Therefore, for these alloy specimens the transition temperature was determined from the temperature variation of the magnetic susceptibility which was measured by dc or ac mutual inductance method. The sample whose heat capacity had been studied were also used to the measurement of the susceptibility. The susceptibility of $Ta_{0.85}Re_{0.15}$ was studied down to 0.5°K using a He^3 cryostat which was a modification of the He^3 calorimeter. The temperature adequate to $Ta_{0.8}Re_{0.2}$ and $Ta_{0.75}Re_{0.25}$ were attained by a magnetic cooling method. The coolant salt and the thermometer salt were made from the ferric ammonium sulphate which was pressed around 200 copper wires of 0.1 mm in diameter. The coolant salt was made in a cylinder form of 10 cm in length and 2 cm in diameter and the thermometer was a spherical salt pill of 1.5 cm in diameter. The sample was attached tightly with GE 7031 adhesive to the copper plate which had been hardsoldered to the middle of the copper wires connecting the coolant with the thermometer pills. The magnetic temperature T^* defined by the susceptibility was determined after White.[8] The lowest temperature attained using ferric ammonium salt was 0.06°K.

§ 3. Experimental Results

A) Superconducting transitions

Figure 3 shows the molar heat capacity divided by temperature C/T versus T^2 plotted for a pure Ta element and nine Ta-Re alloy specimens. Although measurements for Ta were made up to $T^2=46$, plots above $T^2=25$ were omitted to show the superconducting region clearly. Five specimens from pure Ta to $Ta_{0.9}Re_{0.1}$ show the distinct superconducting phase transition in the temperature range studied. Transition temperatures, T_c, for Ta and $Ta_{0.975}Re_{0.025}$ were deduced precisely from a sharp turning point on heating curves in the heat capacity measurement. For $Ta_{0.95}Re_{0.05}$, $Ta_{0.925}Re_{0.075}$, and $Ta_{0.9}Re_{0.1}$, T_c was taken as the midpoint of the transition region of heat capacity, the largest width of which was 0.19°K for $Ta_{0.9}Re_{0.1}$. Since the complete transition to the superconducting state for $Ta_{0.85}Re_{0.15}$ could not be observed in the studied temperature range of heat capacity, the transition temperature was determined by ac mutual inductance method to be the point of half value of the susceptibility in the complete superconducting state. The transition temperature thus determined was 0.75°K and transition width was 0.20°K. While the beginning of the transition was also seen in the plots of the heat capacity. The transition temperature for $Ta_{0.8}Re_{0.2}$ was 0.21°K. The width of its transition could not be well defined, because the measuring field of 0.1 gauss yielded only 2 mm deflection for the detector even in the complete superconducting state. Superconductivity for $Ta_{0.75}Re_{0.25}$ could not be detected down to 0.06°K. Transition temperatures thus determined for seven specimens are summarized in Table I and plotted as a function of electrons per atom ratio, n, in Fig. 4. T_c decreases monotonically with increasing electrons per atom ratio.

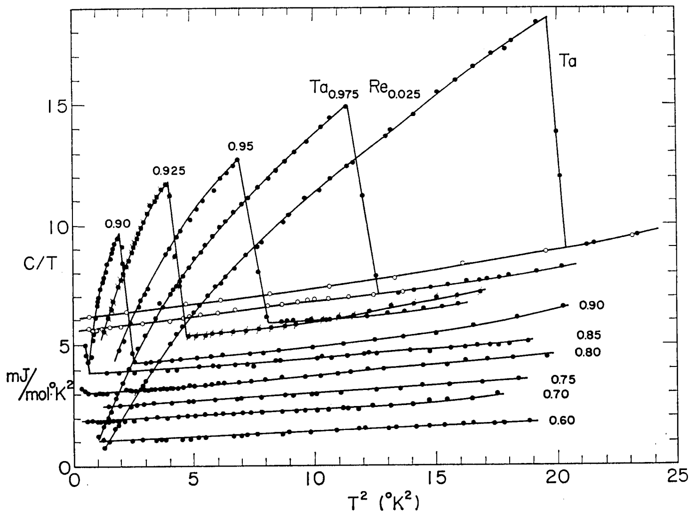

Fig. 3. Molar heat capacity divided by temperature versus square of temperature for Ta-Re alloy series. Atomic fractional concentrations of Ta are shown in the figure.

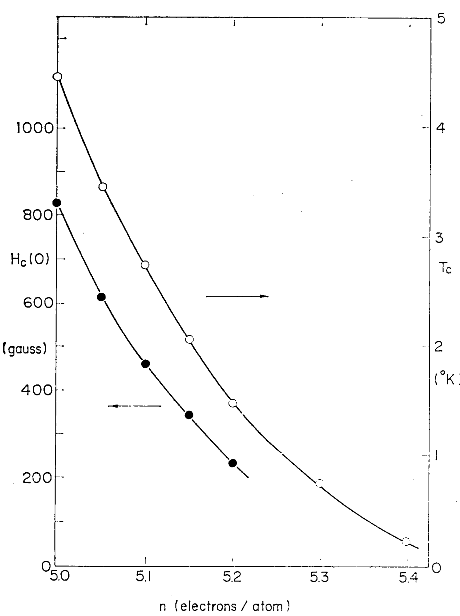

Fig. 4. Critical field at 0°K, $H_c(0)$, and superconducting transition temperature, T_c, versus the number of electrons per atom, n.

B) *Normal state*

A field of about 3000 gauss was used to quench superconductivity for pure Ta and $Ta_{0.975}Re_{0.025}$. These samples showed the transition at rather high temperatures. In the case of specimens whose superconductivity was detected, except Ta and $Ta_{0.975}Re_{0.025}$, the plots in normal state above the transition temperature were used to obtain the parameters of normal state specific heat. The specific heat in the normal state, C_n, can be expressed at low temperatures in the form

$$C_n=\gamma T+\beta T^3 , \qquad (2)$$

where γ and β are coefficients of the electronic and lattice contributions, respectively. The Debye temperature Θ near 0°K can be obtained from the expression

$$\Theta=\left(\frac{1944}{\beta}\right)^{1/3}{}^{\circ}\mathrm{K} , \qquad (3)$$

where β is expressed in J/mol·°K⁴. For each of the alloys, normal portion of C/T versus T^2 curve deviates upwards from a straight line at higher temperature, which follows from the appearance of a T^5 term in C_n. For each of specimens, therefore, the plots in the linear C/T versus T^2 at the lowest temperature region were used to obtain more accurate values of γ and β. The values of γ and Θ are collected in Table I. The estimated errors are ± 0.05 mJ/mol·°K² for γ and ± 7°K for Θ. $\gamma=6.15$ mJ/mol·°K² with $\Theta=250$°K for Ta are in agreement with recent data $\gamma=6.02$ mJ/mol·°K² with $\Theta=258$°K by Shen[7] and $\gamma=6.28$ mJ/mol·°K² with $\Theta=240$°K by Morin and Maita.[9] The values of γ and Θ are plotted as a function of electrons per atom ratio in Fig. 5. γ decreases and Θ increases monotonically with increasing electrons per atom ratio. γ for Ta by Shen[7] and for Ta-W alloys belonging bcc transition series by Bucher *et al.*[1] are also included in Fig. 5 and agree fairly well with the present data.

C) *Superconducting state*

In the case of alloys which showed a complete

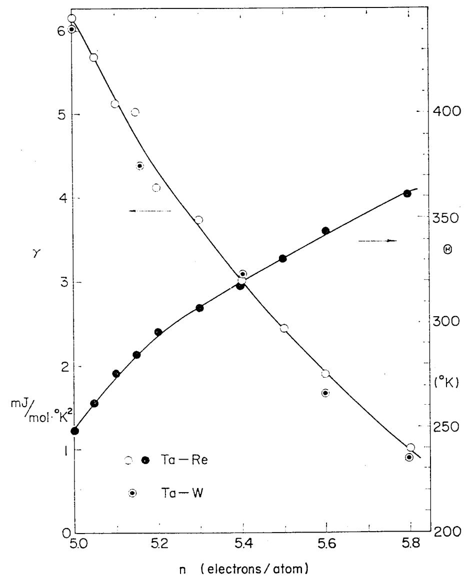

Fig. 5. Electronic specific heat coefficient, γ, and Debye temperature, Θ, versus the number of electrons per atom, n. γ for Ta by Shen and for Ta-W alloys by Muller *et al.* are also shown with double circles (⊙).

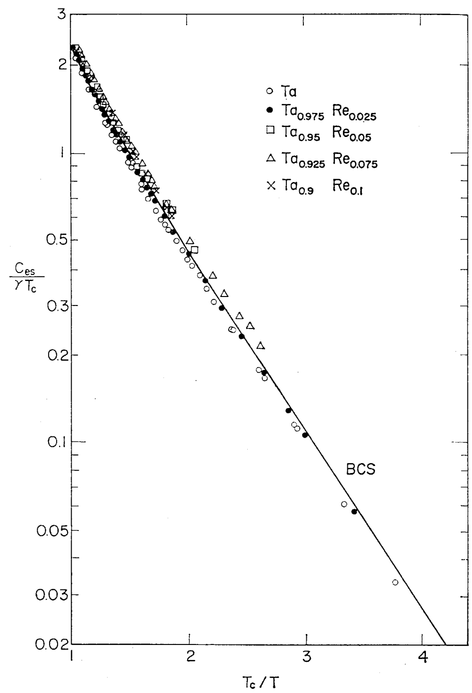

Fig. 6. The reduced superconducting state electronic specific heat, $C_{es}/\gamma T_c$, versus T_c/T. The BCS curve is also shown.

superconducting phase transition in the studied temperature region, the separation of the electronic contribution, C_{es}, to the superconducting specific heat, C_s, was made by assuming that the lattice contributions to C_s and to C_n were equal. The assumption that the lattice contributions to the specific heats of both superconducting and normal states were equal has been ascertained by Shen[7] to be correct in the cases of pure and impure Ta. $C_{es}(T)$ is expressed in the form of

$$C_{es}(T)=C_s(T)-\{C_n(T)-\gamma T\}\ , \qquad (4)$$

where $C_n(T)$ is the measured specific heat of normal state including the T^5 term. $\log_{10}(C_{es}/\gamma T_c)$ is plotted as a function of the reciprocal reduced temperature T_c/T in Fig. 6. From the linear plots of Fig. 6 at $T_c/T>1.5$, C_{es} may be approximated by

$$C_{es}/\gamma T_c=a\exp(-bT_c/T)\ . \qquad (5)$$

The values of a and b for four specimens are listed in Table I. For $Ta_{0.9}Re_{0.1}$ these values were not tabulated because sufficiently large T_c/T was not obtained in this case. However, in the following analysis the values of a and b for this specimen were taken to be equal to the values for $Ta_{0.925}Re_{0.075}$, since the values of a and b were not distinguished between $Ta_{0.9}Re_{0.1}$ and $Ta_{0.925}Re_{0.075}$ in Fig. 6.

The critical field $H_c(T)$ at temperature T could be calculated by the relation:

$$\int_T^{T_c}(S_n-S_s)\mathrm{d}T=\frac{VH_c(T)^2}{8\pi}\ , \qquad (6)$$

where V means molar volume which, for Ta-Re alloys, was taken as appropriate values interpolated between molar volumes of Ta and Re. The entropy difference (S_n-S_s) was found by measuring the appropriate areas under the curve of the $C_n/T-C_s/T$ versus temperature. The value of C_s at lower temperatures is not the measured value but the extrapolated one using eq. (5). For alloys which were not studied in magnetic fields, the value of C_n near absolute zero was determined assuming that the entropies of the two states were equal at the transition temperature. Using the same procedure the values of the electronic specific heat coefficient, γ, for alloys $Ta_{0.95}Re_{0.05}$, $Ta_{0.925}Re_{0.075}$, and $Ta_{0.9}Re_{0.1}$ were also determined. The critical field at 0°K, $H_c(0)$, for Ta was found to be 831 gauss. This is a little larger than the value 824 gauss deduced from magnetic measurement by Hinrichs and Swenson[10] and 814 gauss obtained from calorimetric measurements by Shen.[7] The values of $H_c(0)$ for five specimens were plotted as a function of the electrons per atom ratio in Fig. 4. $H_c(0)$ and T_c decrease as the electrons per atom ratio increases. The temperature dependence of $H_c(T)$ is shown in Fig. 7 as the deviation from a

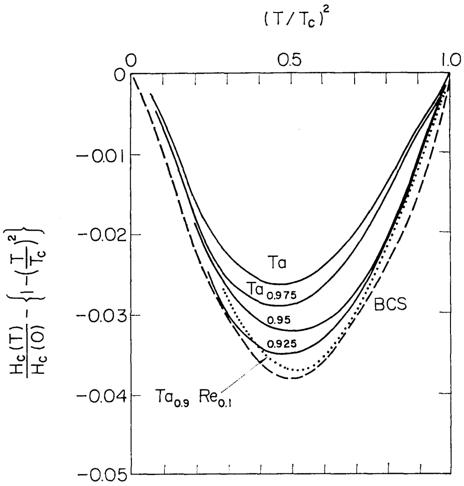

Fig. 7. Deviation of the critical field from a parabolic curve versus $(T/T_c)^2$. Atomic fractional concentrations of Ta are shown in the figure. The BCS result is also shown with a dashed curve.

parabolic curve in the temperature range where the measurements of specific heat were made. The deviation curve of Ta is in good agreement with that obtained by Shen[7] and by Hinrichs and Swenson.[10] Figure 7 reveals that as the Re concentration increases, the deviation curve approaches to that predicted by the Bardeen-Cooper-Schrieffer[11] (BCS) theory of superconductivity or the weak coupling limit. This result is consistent with the tendency of decreasing of T_c/Θ. This tendency of approaching to the BCS limit with increasing Re concentration is also seen in the tendency of the superconducting energy gap $2\varepsilon_0/kT_c$ and the values of b.

According to the BCS theory[11] the values of the superconducting energy gap $2\varepsilon_0$ at 0°K is given by the following relation:

$$2\varepsilon_0/kT_c=[2\pi VH_c(0)^2/3\gamma T_c]^{1/2}\ . \tag{7}$$

Meanwhile, Toxen[12] found out empirically that the energy gap at 0°K was closely related to the slope of the critical field at the transition temperature as follows:

$$\frac{2\varepsilon_0}{kT_c}=\left|\frac{2T_c}{H_c(0)}\cdot\frac{\mathrm{d}H_c}{\mathrm{d}T}\right|_{T=T_c}. \tag{8}$$

The energy gaps at 0°K obtained from both eqs. (7) and (8) are in excellent agreement each other and are summarized in Table I. In Fig. 8 the

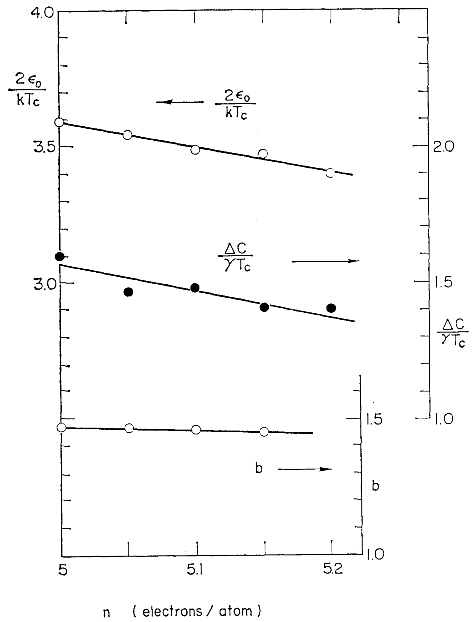

Fig. 8. The reduced superconducting energy gap at 0°K, $2\varepsilon_0/kT_c$, the reduced jump in specific heat, $\Delta C/\gamma T_c$, and the exponent of the reduced superconducting state electronic specific heat, b, versus the number of electrons per atom, n.

values of $2\varepsilon_0/kT_c$ deduced from eq. (7) are plotted against the electrons per atom ratio together with the value of b in eq. (5). The latter value is a measure of the energy gap. The tendency of the decrease of b with increasing electrons per atom ratio is in accordance with the decrease of $2\varepsilon_0/kT_c$.

If $\Delta C=C_s-C_n$ at $T=T_c$ is defined to be a jump in zero field superconducting transition, the jump can be expressed in terms of the slope of the critical field from the consideration of the second order phase transition as follows:

$$\frac{\Delta C}{\gamma T_c}=\frac{1}{\gamma T_c}\,\frac{T_c}{4\pi}\left(\frac{\mathrm{d}H_c}{\mathrm{d}T}\right)^2_{T=T_c}. \tag{9}$$

The slopes of the critical field at the transition temperature and the specific heat jumps deduced from eq. (9) are summarized in Table I together with the directly measured specific heat jumps, which are in good agreement with the calculated ones. The measured specific heat jumps are plotted as a function of electrons per atom ratio in Fig. 8. According to the BCS theory,[11] the ratio of the jump to γT_c is given by

$$\frac{\Delta C}{\gamma T_c}=-\frac{1}{\gamma T_c}\cdot N(0)\cdot\left(\frac{\mathrm{d}\varepsilon_0(T)^2}{\mathrm{d}T}\right)_{T=T_c}, \tag{10}$$

where $\varepsilon_0(T)$ indicates the superconducting energy gap as a function of temperature. Assuming the corresponding rule for the energy gap, *i.e.* $\varepsilon_0(T)=\varepsilon_0(0)f(T/T_c)$, one can obtain the following relation:

$$\frac{\Delta C}{\gamma T_c}=-k^2\frac{N(0)}{\gamma}\left[\frac{\varepsilon_0(0)}{kT_c}\right]^2\frac{\mathrm{d}f(t)^2}{\mathrm{d}t}\bigg|_{t=1}, \quad (11)$$

where $f(t)$ is a universal function for alloys. Considering that the density of states at Fermi energy $N(0)$ is proportional to the value of γ, one can see that $\Delta C/\gamma T_c$ is a function of $[\varepsilon_0(0)/kT_c]^2$ alone. The values of $\Delta C/\gamma T_c$ and $2\varepsilon_0/kT_c$ plotted in Fig. 8 against the electrons per atom ratio show that $\Delta C/\gamma T_c$ increases with increasing $2\varepsilon_0(0)/kT_c$, which is consistent with the tendency of eq. (11).

§ 4. Discussion

In this chapter we calculate the Coulomb interaction and the electron phonon interaction parameters from the present experimental results and discuss superconducting properties and electronic density of states of Ta-Re alloys. Recently, McMillan[2] has given a theory of the transition temperature of strong-coupled superconductors. Assuming the appropriate phonon spectrum of density of states, he has written down the integral equations for the strong-coupled superconductors. He has presented accurate numerical solutions of the integral equations by an interaction method and shown that T_c was given as a function of the electron-phonon and Coulomb interaction parameters λ and μ^*, respectively:

$$T_c=\frac{\Theta}{1.45}\exp\left[-\frac{1.04(1+\lambda)}{\lambda-\mu^*(1+0.62\lambda)}\right]. \quad (12)$$

This relation holds well for superconductors of weak and intermediate coupling $\lambda<1$, while for the strong-coupled superconductors the appropriate condition is required. The value of γ is proportional to the density of states at the Fermi surface which is called the bare or band structure density of states. On the other hand γ is enhancd by the electron phonon interaction and is written in terms of the bare density of states $N_0(0)$:

$$\gamma=\frac{2\pi^2k^2}{3}N_0(0)(1+\lambda). \quad (13)$$

Therefore if λ and γ are known, one can find out $N_0(0)$. McMillan has determined the value of μ^* empirically from the isotope shift. The value is about 0.13 which is the average for several transition metals. On the other hand one can derive μ^* from the measured electronic specific heat using the expression of Morel and Anderson.[13]

$$\mu^*=\frac{\mu}{1+\mu\ln(E_B/\omega_0)}, \quad (14)$$

where E_B is the electronic bandwidth and ω_0 is the maximum phonon frequency. Using a Fermi-Thomas model μ was given by

$$\left.\begin{aligned}\mu&=\frac{1}{2}a^2\ln(1+1/a^2),\\ a^2&=\frac{3e^2\hbar^2\gamma}{4\pi k^2m^*E_F},\end{aligned}\right\} \quad (15)$$

where E_F is the Fermi energy and m^* is the effective mass of the electron *i. e.* $m^*=(1+\lambda)m_0$. For Ta-Re alloy series, E_F is 6.8 to 7.4 eV according to theoretical band structure for W obtained by Mattheiss.[3] Using eqs. (12)–(15), one should obtain λ and μ^* in principle. However, it is very hard to solve these equations. Therefore, one makes an approximation as follows. In order to obtain μ^* for Ta-Re series we must determine m^* first, using λ which was obtained from McMillan's value $\mu^*=0.13$ as a first approximation. Then putting this m^*, the measured value of γ, and the above-descrived E_F into eqs. (14) and (15), the new value of μ^* for Ta was determined to be 0.111. As the Re concentration increases, the value of μ^* decreases very slightly and becomes to 0.107 for $Ta_{0.8}Re_{0.2}$. Accordingly, μ^* was taken to be 0.11 over the Ta-Re alloy series. Using $\mu^*=0.11$, the experimentally determined transition temperature T_c, and Debye temperature Θ, eq. (12) gives the phonon coupling constant λ. Putting these γ and λ into eq. (13) the bare density of states $N_0(0)$ can be derived. The values of λ and $N_0(0)$ thus determined are summarized in Table I and plotted against electrons per atom ratio in Figs. 9 and 10 together with other author's[1] data on bcc 5d transition alloys. λ behaves, as a function of electrons per atom ratio, in the same way as $N_0(0)$. λ is plotted in Fig. 11 against $N_0(0)$ which turns out to be proportional to λ and is expressed as

$$\lambda=0.72N_0(0). \quad (16)$$

Therefore the behavior of λ is the direct reflection of $N_0(0)$. If one assumes the linear relation between λ and $N_0(0)$, λ values for alloys which did not show superconductivity, can be calculated from the measured value of γ. By combining eqs. (13) and (16), λ is given by

$$\lambda=\frac{-1+\sqrt{(4.32/\pi^2k^2)\gamma+1}}{2}. \quad (17)$$

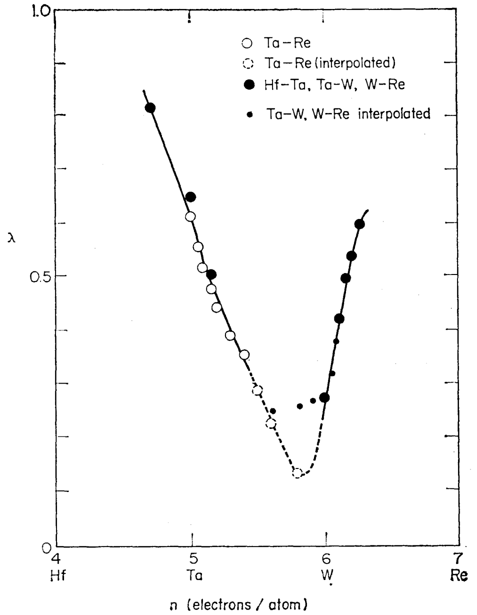

Fig. 9. The electron-phonon coupling constant, λ, versus the number of electrons per atom, n, for bcc 5d transition metal alloys. Present data for Ta-Re alloys were deduced with the Coulomb interaction $\mu^*=0.11$. Data for Hf-Ta, Ta-W, and W-Re alloys were obtained with $\mu^*=0.13$ by McMillan.

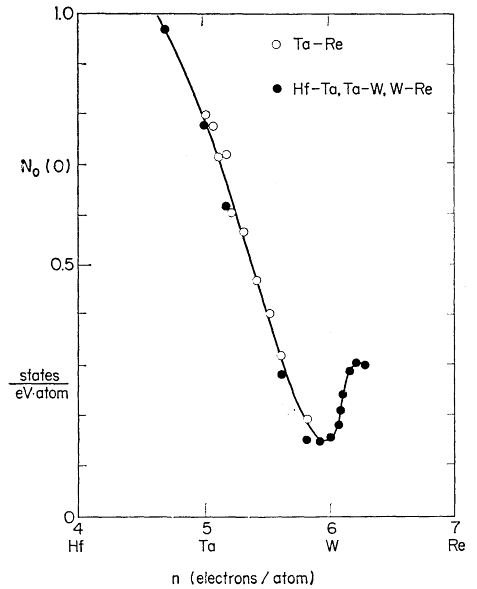

Fig. 10. The bare density of states, $N_0(0)$, versus the number of electrons per atom, n, for bcc 5d transition metal alloys.

For $Ta_{0.75}Re_{0.25}$, $Ta_{0.7}Re_{0.3}$, and $Ta_{0.6}Re_{0.4}$, the values of λ which are obtained using the above procedure are listed with parentheses in Table I.

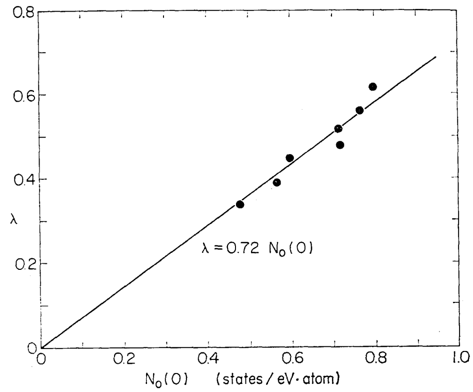

Fig. 11. The electron phonon coupling constant, λ, versus the bare density of state, $N_0(0)$, for Ta-Re alloys. The straight line shows the relationship $\lambda=0.72N_0(0)$.

Using these values of λ and eq. (12) one can obtain the transition temperatures of 0.04°, 0.002° and 10^{-36}°K for $Ta_{0.75}Re_{0.25}$, $Ta_{0.7}Re_{0.3}$, and $Ta_{0.6}Re_{0.4}$, respectively. The extreme low transition temperature for $Ta_{0.6}Re_{0.4}$ may not be plausible, considering that transition temperatures for 4d series Nb-Mo alloys[14] are not so low. If this is not the real fact, the reason why the calculated value of T_c for $Ta_{0.6}Re_{0.4}$ is extremely low may be that at small λ values $N_0(0)$ is not always proportional to λ and moreover Coulomb interaction μ^* can not be taken to be $\mu^*=0.11$.

Next we discuss the density of states of Ta-Re alloys. Mattheiss has calculated the band structures of Ta and W[3] using the augmented plane wave (APW) method including relativistic effects for Ta and nonrelativistically for W. The potentials he used were derived from superposed atomic charge densities which were obtained from self-consistent calculations involving a $(5d)^3(6s)^2$ atomic configuration for Ta and $(5d)^5(6s)^1$ for W, respectively. The calculation of the energy band was carried out at uniformly distributed points in the Brillouin zone for Ta or W. These results were converted to the density of states by a method of a weighted average of the APW eigenvalues or an interpolation scheme. The theoretical density of states for electrons has been compared by McMillan[2] with the experimental density of states for bcc 5d transition metal alloys. However, the experimental density of states he used is not complete enough but included the interpolated plots for several alloys of bcc 5d transition metal (Ta-W and W-Re) whose superconductivity was not detected. The theoretical density of states for Ta according to Mattheiss and the pre-

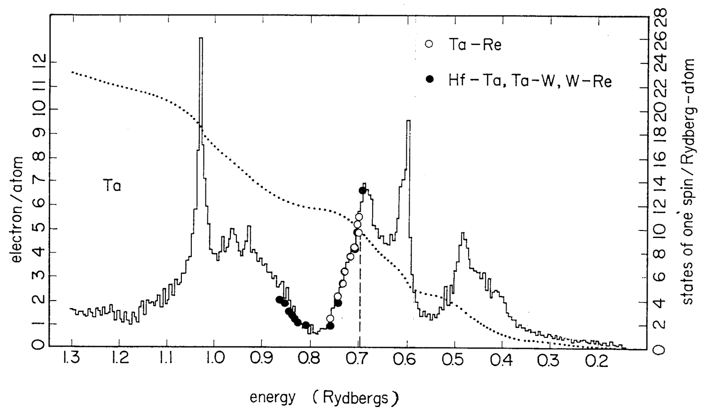

Fig. 12. The theoretical bare density of states for Ta by Mattheiss and the experimental data for bcc 5d transition metal alloys. Open circles (○) show the present data of Ta-Re and solid circles (●) indicate the data of Hf-Ta, Ta-W, and W-Re, respectively, including the interpolated data. (1 Rydberg≡1 eV).

sent result of Ta-Re alloys are shown in Fig. 12 together with other author's data of Hf-Ta, Ta-W, and W-Re alloys. For $Ta_{0.75}Re_{0.25}$, $Ta_{0.7}Re_{0.3}$, and $Ta_{0.6}Re_{0.4}$ the interpolated $N_0(0)$'s are plotted in Fig. 12. The theoretical and experimental densities of states for Ta-Re alloys (Ta rich side) are in good agreement, but those for W and W-Re alloy (W rich side) are less than the theoretical density of states in Fig. 12. On the contrary, the experimental density of states for Ta-Re alloys is 10 to 20% larger than the theoretical density calculated for W, and data for W and W-Re alloys are in good agreement with the theoretical one calculated for W (Fig. 10 in ref. 2)). It is concluded that data for Ta rich alloys agree well with the band calculation for Ta and data for W rich alloys show agreement with the W calculation. When one proceeds from Ta to W, a rigid band density of states model can not be applied for both elements in general, but some deviations from the model can be found in the cases of Ta and W. As Mattheiss pointed, the interpolation scheme yields crude density of states because the number of points in the Brillouin zone are limited and therefore some of peaks in the density of states may be overlooked. Meanwhile, more accurate values of the Fermi energy and the density of states at the Fermi energy can be obtained in such a way that the square of the wave vector are expanded in lattice harmonics. According to Mattheiss, the accurate density of states for Ta, 0.65 states/eV·atom does not agree with the experimental values of 0.77, but crude density of states of about 0.80 states/eV·atom agrees better. It is interesting that the experimental data not only for Ta but also for Ta-Re alloy series are in good agreement with the crude density of states.

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